

Sunan Zou

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Personal Profile

I am a fourth-year Ph.D. candidate in the School of Computer Science at Peking University and a member of the PKU-DASYS Lab at the Center for Energy-Efficient Computing and Applications (CECA), advised by Prof. Guojie Luo. I received my B.S. in Computer Science and Technology from Peking University in 2022. My research interests include LLM systems and infrastructure, domain-specific accelerator compilation frameworks, and AI compilers. I currently focus on kernel-schedule co-design for emerging model architectures, as well as model compression and compression-aware operator/architecture design. I also have extensive experience in electronic design automation, particularly logic synthesis optimization. My doctoral research centers on system-level applications of Sketch algorithms, hardware optimization, and automated accelerator generation.

Education

Ph.D. in Computer Architecture

School of Computer Science, Peking University
Advisor: Prof. Guojie Luo

Sep 2022 – Jun 2027 (expected)

Beijing, China

B.S. in Computer Science and Technology

School of Electronics Engineering and Computer Science (EECS), Peking University

Sep 2018 – Jul 2022

Beijing, China

Research Experience

Dots, Rednote

Research Intern, Ace Talent Program.

Topics: xHC Fusion and Scheduling for LLM Training Systems

May 2026 - Now

Beijing, China

Yamagishi Lab, National Institute of Informatics (Japan)

Research Intern, advised by Prof. Junichi Yamagishi.

Topics: Neural-Language Model for Violin Synthesis

Oct 2025 - Apr 2026

Tokyo, Japan

Electrical and Computer Engineering, Carnegie Mellon University

Research Intern, advised by Prof. Larry Pileggi.

Topics: Graph Neural Network-Based Logic Locking Detection

Mar 2021 - Sep 2021

Remote

DASYS Lab, Peking University

Research Intern, advised by Prof. Guojie Luo.

Topics: Reinforcement Learning-Based Logic Optimization

Jul 2019 - Jan 2022

Beijing, China

Publications

- [C9] **Sunan Zou**, Zhe Zhang, Yigitcan Özer, Guojie Luo, and Junichi Yamagishi. “VioLM: A Neural Language Model for Violin Synthesis with Articulation”. *The 27th International Society for Music Information Retrieval Conference (ISMIR)*, 2026. (To Appear)
- [C8] Xiao Tan, Chenyue Li, **Sunan Zou**, and Guojie Luo. “RTLPilot: Skill-Driven Multi-Agent System for Repository-Level RTL Code Migration”. *2026 International Symposium of EDA (ISED)*, 2026. (To Appear)
- [C7] **Sunan Zou**, Xueting Sun, Ziyun Zhang and Guojie Luo. “UltraSketchLLM: Sub-1-Bit LLM Compression via Sketch and Hardware-Friendly Operators”. *63st Design Automation Conference (DAC)*, 2026.
- [C6] **Sunan Zou**, Bizhao Shi, Ziyun Zhang and Guojie Luo. “MuSA: Multi-Sketch Accelerator with Hybrid Parallelism and Coalesced Memory Organization”. *IEEE 42nd International Conference on Computer Design (ICCD)*, 2024.

- [C5] Xinming Wei, Ziyun Zhang, **Sunan Zou**, Kaiwen Sun, Jiahao Zhang, Jiaxi Zhang, Ping Fan, and Guojie Luo. "AceRoute: Adaptive Compute-Efficient FPGA Routing with Pluggable Intra-Connection Bidirectional Exploration". *43th IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2024.
- [C4] Bizhao Shi, Tuo Dai, **Sunan Zou**, Xinming Wei, and Guojie Luo. "ImageMap: Enabling Efficient Mapping from Image Processing DSL to CGRA". *30th International European Conference on Parallel and Distributed Computing (Euro-Par)*, 2024.
- [C3] **Sunan Zou**, and Guojie Luo. "PONO: Power Optimization with Near Optimal SMT-based Sub-circuit Generation". *61st Design Automation Conference (DAC)*, 2024.
- [C2] **Sunan Zou**, Jiaxi Zhang, and Guojie Luo. "Incremental SAT-based Exact Synthesis", *34th Great Lakes Symposium on VLSI (GLSVLSI)* 2024. **[Best Paper Reward]**
- [C1] **Sunan Zou**, Jiaxi Zhang, Bizhao Shi, and Guojie Luo. "BESWAC: Boosting Exact Synthesis via Wiser SAT Solver Call", *27th Design Automation and Test in Europe (DATE)*, 2024.
- [J2] Lei Chen, Yiqi Chen, Zhufei Chu, Wenji Fang, Tsung-Yi Ho, et al, and **Sunan Zou (Alphabetical Order)**. "Large circuit models: opportunities and challenges". *Science China Information Sciences Volume 67*, 2024.
- [J1] **Sunan Zou**, Jiaxi Zhang, Bizhao Shi, and Guojie Luo. "PowerSyn: A logic synthesis framework with early power optimization", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems Volume: 43 (TCAD)*, 2023.
- [I1] Xiangdong Zhang, Xiaohan Qin, **Sunan Zou**, Tuo Dai, et al. "xHC: Expanded Hyper-Connections". *A Technical Report (arXiv preprint, arXiv:2607.14530)*, 2026.

Representative Projects

[LLM Training Systems] xHC Fusion & Communication-Aware Scheduling

May 2026 - Now

Contents: Optimized the fine-grained computation and multi-stream state movement introduced by xHC in LLM training within Megatron Core. Designed GPU kernels tailored to xHC tensor shapes and memory-access patterns, and selected fusion boundaries based on data dependencies, data reuse, and GPU execution characteristics. Co-designed Triton forward/backward implementations and custom Autograd paths so activation saving, recomputation, and sparse state materialization align with the fusion boundaries. At the scheduling level, explored shared control computation across sublayers and state management under CUDA Graph and combined IF1B execution; further explored pre-communicating inactive streams across PP/VPP boundaries to overlap communication with active-path computation. Validated correctness and performance through numerical checks, profiling, and end-to-end training.

Role: Main contributor. **Publication:** [I1]

[LLM Algorithm] Violin Audio Model

Oct 2025 - Apr 2026

Contents: This project focuses on the design of a two-stage transfer training framework and data generation paradigm for musical instrument audio generation. First, the project proposes a score tokenization method that accounts for playing techniques, upon which a data generation paradigm is constructed comprising score legalization, dynamic fitting sampling, and automatic articulation annotation, addressing the data scarcity problem. Building on full-parameter transfer training of the language model using generated data, a small amount of real data is employed for joint partial fine-tuning of both the language model and the encoder model to improve audio generation fidelity. This project was conducted as a research topic during an exchange visit at the National Institute of Informatics (NII), Japan, with a planned submission to ACM Multimedia 2026.

Role: Independent project. **Publication:** [C9]

[SW/HW Co-design] Sketch-Based LLM Compression & Hardware-Friendly Kernel

May 2025 - Feb 2026

Contents: This project applies Sketch algorithms to compress LLM weights and develops fused compression and decompression kernels to reduce associated overhead. We first compress model weights using a Sketch algorithm tailored for LLMs, achieving sub-1-bit compression rates while maintaining model performance. To reduce the inference overhead and training performance loss introduced by decompression, we reformulate the sequential

hashing operations in Sketch as equivalent matrix multiplication operations, and develop fused kernels that integrate compression/decompression with computation, effectively reducing additional overhead.

Role: Independent project. **Publication:** [C7]

[Compiler] Sketch Accelerator Design & Auto-Generation

Feb 2024 - Now

Contents: This project focuses on the design of an accelerator generation and optimization framework adaptable to multiple Sketch algorithms and diverse backend hardware targets. First, we define an extensible domain-specific language (DSL) capable of expressing hierarchical and multi-Sketch application scenarios and algorithm designs. Second, we propose efficient analytical models to estimate key metrics, providing guidance for subsequent architecture DSE. The project spans multiple backend targets including FPGAs, GPUs, and CPUs. The framework will be integrated with system-level applications of various Sketch algorithms, including network monitoring, hot-cold page detection, and working set estimation, to optimize overall runtime efficiency. Follow-up work from this project is planned for submission to ASPLOS 2027.

Role: Independent project. **Publication:** [C6]

[Design Automation] Multi-Level Parallel FPGA Routing

Oct 2023 - Mar 2024

Contents: This project focuses on significantly accelerating FPGA routing search through a combined coarse- and fine-grained parallelization approach. On one hand, the project employs inter-signal parallelism by recursively partitioning parallel regions and leveraging directed acyclic graphs (DAGs) to resolve potential inter-signal conflicts. On the other hand, we introduce bidirectional parallelism in the search process from source nodes to sink nodes, which, combined with inter-signal parallelism, further improves overall search efficiency.

Role: Main contributor & Code maintainer. **Publication:** [C5]

[Design Automation] SAT-Baed Incremental Synthesis Acceleration

Jan 2023 - Jan 2025

Contents: This project focuses on reducing the runtime and increasing the solvable problem scale of exact synthesis. Upon observing that different Boolean circuits are better suited to different encoding schemes, this project employs machine learning methods to predict appropriate encodings from Boolean function features, thereby improving SAT solving efficiency. The project further exploits the optimal substructure property present in certain optimal circuits, designing heuristic auxiliary functions to extract subproblems from the target problem, and developing two incremental SAT solving strategies based on topology and a sub-circuit. Follow-up work will develop a problem-specific SAT solver, with a planned submission to TCAD.

Role: Independent project. **Publication:** [C1, C2]

[Design Automation] Reinforcement Learning for Logic Optimization

Jul 2021 - Jul 2023

Contents: This project formulates the logic synthesis optimization command search problem as a reinforcement learning task, and optimizes circuit designs by incorporating an efficient power modeling method as the reinforcement learning reward function. First, the project extends the exact synthesis optimization objective using SMT formulations with a power modeling approach that accounts for glitches and dynamic signals, generating locally optimal low-power circuits and integrating them into conventional logic optimization commands. Second, the project manually engineers circuit representation features as the reinforcement learning environment, trains a command decision model with power, area, and performance as joint optimization objectives, and produces circuit-specific optimization sequences that effectively improve the power characteristics of logic synthesis without compromising area or performance.

Role: Independent project. **Publication:** [J1, C3]

Honors and Awards

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| • Honors for Merit Student (三好学生), Peking University | 2024 |
| • BYD Scholarship (北京大学比亚迪奖学金), Peking University | 2024 |
| • ISFPGA FPGA Routing Contest Third Place, ISFPGA Committee | 2024 |
| • Principal's Fund for Undergraduate Research Training (Natural Science), Peking University | 2021 |
| • Freshman Scholarship (新生奖学金), Peking University | 2018 |
| • Freshman Scholarship (院长新生奖学金), School of EECS, Peking University | 2018 |

Services

- **Official Reviewer:** TCAD
- **Teaching Assistant:** Introduction to Parallel and Distributed Computing, Peking University 2022, 2023

Skills

- **Programming Languages:** C/C++, Python(Pytorch), FPGA HLS, Verilog, Triton, CUDA
- **Tools:** LATEX, Git, Docker/Apptainer
- **Languages:** Chinese, English, and Japanese.
- Former Member of Peking University Student Orchestra